

An operator range in ℓ_1 that is not an injective operator range

Candidate full-counterexample packet

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Status. Candidate full counterexample; likely valid; human review recommended.

1 Source question

Rosenthal and Troitsky use the following finite-power definition. A linear subspace Y of a Banach space X is an *operator range* if $Y = T(M)$ for a bounded operator $T : M \rightarrow X$, where M is a closed subspace of X^n for some finite n . It is an *injective operator range* if such a presentation can be chosen with T one-to-one. Remark 4.4 of [1] asks whether every operator range in a Banach space is injective in this sense. The exact source passage (PDF page 13) is reproduced below.

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Remark 4.4. It is easily seen that every operator range in a Hilbert space is injective of order one. It seems we need the concept of operator range of order n , however, to obtain a general result. Also, we don't know if every operator range in a Banach space is also an injective operator range. This question was posed by P. Rosenthal.

Before proving the theorem, we need some preliminary results, which also hold in real Banach spaces. We start with the following simple consequences of the definition of an operator range, given in [Foi72] in a Hilbert space case.

The answer is no, already for an order-one range in $X = \ell_1$.

2 Proof intuition

The construction separates the Banach space used to *generate* a range from the ambient geometry allowed in an injective presentation. The space ℓ_2 is a quotient of ℓ_1 , and it admits a bounded one-to-one diagonal map J into ℓ_1 . Composing a quotient $Q : \ell_1 \rightarrow \ell_2$ with J therefore makes $J(\ell_2)$ an operator range in ℓ_1 .

If the same set were the range of an injective operator from a closed subspace N of a finite power of ℓ_1 , the two parametrizations would give a linear bijection $N \rightarrow \ell_2$. Although the inverse of J is not continuous on its range with the ambient ℓ_1 norm, the equality of the two parametrizations makes this bijection closed. The closed graph and open mapping theorems then force $N \cong \ell_2$. This is impossible: every closed subspace of a finite power of ℓ_1 has the Schur property, whereas ℓ_2 does not.

3 A general quotient–embedding obstruction

Lemma 1. *Let X and Z be Banach spaces. Suppose that for some finite r there are a closed subspace $M \subseteq X^r$, a bounded surjection $Q : M \rightarrow Z$, and a bounded injection $J : Z \rightarrow X$. Put $Y = J(Z)$. Then Y is an operator range in X .*

If Y is an injective operator range, then Z is isomorphic to a closed subspace of X^s for some finite s .

Proof. The bounded map $JQ : M \rightarrow X$ has range $J(Z) = Y$, proving the first assertion.

For the second, suppose that $N \subseteq X^s$ is closed and $S : N \rightarrow X$ is bounded and one-to-one with $S(N) = Y$. Define

$$U = J^{-1}S : N \longrightarrow Z.$$

This is a well-defined linear bijection. It has closed graph. Indeed, if $u_k \rightarrow u$ in N and $Uu_k \rightarrow z$ in Z , then continuity of S and J gives

$$Su = \lim_k Su_k = \lim_k J(Uu_k) = Jz.$$

Since J is injective, $Uu = z$. The closed graph theorem makes U bounded, and the open mapping theorem makes U^{-1} bounded. Hence $N \cong Z$. $\square$

4 The ℓ_1 counterexample

We include a construction of the standard quotient needed below.

Lemma 2. *There is a bounded linear surjection $Q : \ell_1 \rightarrow \ell_2$.*

Proof. Choose a sequence (u_k) in the unit sphere of ℓ_2 such that every tail is dense, and define

$$Q(a) = \sum_{k=1}^{\infty} a_k u_k \quad (a \in \ell_1).$$

Then $\|Q\| \leq 1$. Given $x \in \ell_2$, set $r_0 = x$. Recursively, whenever $r_{j-1} \neq 0$, choose a strictly increasing index k_j so that

$$\left\| \frac{r_{j-1}}{\|r_{j-1}\|_2} - u_{k_j} \right\|_2 < \frac{1}{2}, \quad r_j = r_{j-1} - \|r_{j-1}\|_2 u_{k_j}.$$

Thus $\|r_j\|_2 < 2^{-j}\|x\|_2$ and $\sum_j \|r_{j-1}\|_2 \leq 2\|x\|_2$. The vector $a \in \ell_1$ with $a_{k_j} = \|r_{j-1}\|_2$ and all other coordinates zero therefore satisfies $Q(a) = x$. $\square$

Theorem 1. *There is a bounded operator $T : \ell_1 \rightarrow \ell_1$ whose range is not an injective operator range of any finite order in the sense of [1]. Consequently, the question in Remark 4.4 of [1] has a negative answer.*

Proof. Let $Q : \ell_1 \rightarrow \ell_2$ be as in Lemma 2, and define the bounded injection

$$J : \ell_2 \longrightarrow \ell_1, \quad J(x_1, x_2, \dots) = (2^{-1}x_1, 2^{-2}x_2, \dots).$$

Boundedness follows from Cauchy–Schwarz:

$$\|Jx\|_1 \leq \left(\sum_{k=1}^{\infty} 4^{-k} \right)^{1/2} \|x\|_2.$$

Put $T = JQ$ and $Y = T(\ell_1) = J(\ell_2)$. Hence Y is an order-one operator range in ℓ_1 .

If Y were an injective operator range, Lemma 1 would make ℓ_2 isomorphic to a closed subspace N of $(\ell_1)^s$ for some finite s . Finite products of ℓ_1 have the Schur property, and that property passes to closed subspaces and is preserved under Banach-space isomorphisms. But ℓ_2 does not have the Schur property: its unit vectors converge weakly to zero and not in norm. This contradiction proves the theorem. $\square$

5 Verification, scope, and literature boundary

The counterexample is stronger than merely requiring a complicated initial domain: Y is the range of the endomorphism $T = JQ$ of ℓ_1 , yet it has no injective presentation from a closed subspace of *any* finite power of ℓ_1 . The proof applies over either the real or complex field.

It is important not to replace the source’s definition by the more common one allowing an arbitrary Banach domain. Under that broader definition one can always quotient the original domain by the kernel and obtain an injective presentation. The finite-power restriction is exactly what the Schur obstruction defeats.

A bounded novelty search on 11 August 2026 used the exact source sentence, “injective operator range” with the source definition, ℓ_1/ℓ_2 combinations, and the earlier operator-range work of Cross, Ostrovskii, and Shevchik [2]. It found literature on arbitrary-domain operator ranges and injective endomorphism ranges, but no exact finite-power-domain counterexample matching Theorem 1. Because terminology varies across this literature and the 1995 paper is extensive, novelty confidence is moderate, not definitive.

Human-review recommendation. Check the definition in Remark 4.4 against the quantification over finite orders. Then audit the closed-graph step in Lemma 1; it is the only point at which the unbounded inverse of J might superficially appear problematic, and the displayed sequential argument resolves it.

References

- [1] H. P. Rosenthal and V. G. Troitsky, *Strictly semi-transitive operator algebras*, J. Operator Theory **53** (2005), 315–329; arXiv:math/0309014.
- [2] R. W. Cross, M. I. Ostrovskii, and V. V. Shevchik, *Operator ranges in Banach spaces, I*, Math. Nachr. **173** (1995), 91–114.